



Al-Balloti Hospital: a DBMS Hospital Project

ABSTRACT

A hospital's environment is an interconnected network of relations and real-life interactions, which would make it convenient to have a system that can help the employees of a hospital to interact with each other without the use of physical means like speakers and hand written reports and discharge notes. for this project we aim to implement a DBMS for a small hospital to further increase the productivity of the workplace.

DBMS

Al-Balloti Hospital: a DBMS Hospital Project

Database class project, academic year 2023 /2024

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-----This section is intended for the Instructor-----

<u>Topic</u>	<u>Mark</u>
Project Requirements and Modeling	
Correctness of Database mapping	
Functional Dependency and Normalization	
Project Tools	
Project Discussion	
Project Completeness	
Project Output Results or reporting (JasperReport, charts, graphs, etc.)	
Project Administration and Management	
Project Report	
Project Idea	
Project Complexity	
Team work	

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Introduction

Al-Balloti Hospital represents any small Hospital system that can be found in the countryside, The hospital has employees like doctors, nurses and lab technicians, the employees interact with each other frequently (even for a small hospital) to do the absolute best to take care of patients.

Even though the employees do their best at their work, there might be set backs and obstacles caused by the inefficient and poor communication among them, which can hinder their abilities to cure patients in dire need, this includes but is not limited to: writing hand written reports and discharge notes which take time and effort to hand them to the correct person or ordering nurses for tasks via speakers and verbally, hence, it's ought to have a computerized system that can lighten the load of physical communication between them.

A computerized solution for the tasks can be a major upgrade for such system, i.e.: nurses can always look back at the order if they forgot and don't have to go back to the doctor, the patient gets a readable and comprehensive discharge note almost immediately, and the doctors can review the past diagnosis of patients at any time without referring to other physicians. Needless to say, all of this will result in a significant reduction of human made errors.

This report will discuss various aspects of the system, starting with the data collection from a small hospital and how all of the information gathered led to an interconnected relational database scheme, and how a graphical user interface was created to interact with the database established.

Project Requirements

Upon investigation, these were the requirements for a database system of the hospital.

- The database **keeps track of 3 types of entities**, **employees**, **patients**, **departments**, **samples** and **diagnosis**.

- The database **keeps track of 3 types of employees**, **doctors**, **nurses** and **lab technicians**, each employee must have a **name**, **birthdate**, **sex**, **phone number**, **city**, **employee ID**, **password**, **shift** and **salary**.
for **each doctor** we store their **specialty**.
- we keep track of a patient's **ID**, **name**, **phone number**, **address**, **sex**, **date of birth**, **blood type**, **admission date**, **nurse note** and **discharge note**.
- We store each department's **name**, **number of available beds** and **floor in the hospital**, each department has **6 beds** and **two nurses** at any given time, if a department is **full**, **no new patient** can be admitted into it.
- When a patient is admitted into a department, he's also assigned to **one** nurse in said department and he is treated by **many** doctors during his stay.
- a doctor can **view the patients in every department**, and can diagnose the patient, this diagnosis will be stored in the database and **can be viewed by other doctors**, after diagnosis the doctor can either **discharge** the patient, **keep him** for further examination, or **order tests** from the nurse assigned to the patient.
- a nurse **receives orders from different doctors**, depending on the nature of tests required the nurse can **either do them** and **give the results to the doctor**, or **give them to the lab technicians** to preform further analysis, a nurse **can't** diagnose a patient, **but can view previous diagnosis** for a patient.
- The lab technician will give the final results to the doctor as a document.
- After the doctor gets the tests back, he decides what's next for the patient (keep for further tests, order new tests, or discharge him).

Functional Dependencies

- Department Table:

	dep_name	floor	no_of_beds
1	Emergency	G	6
2	Surgical	1st	6
3	Pediatrics	G	6
4	Gynecology	1st	6
5	Cardiology	2nd	6
6	ICU	1st	6
7	Laboratory	2nd	6

The Functional Dependencies are as follows:

- dep_name -> floor

- dep_name -> no_of_beds

- Patient Table:

	1	2	3	4	5	6	7	8	9
id_no	5000	5002	5001	5005	5003	5004	5010	5011	5111
fname	Hashem	Shehdeh	Naeem	Jolia	Vector	Hothaifah	qwer	qwer	qwer
mname	Muthaffar	Jodi	Samir	abo	Jaden	abo	wqer	wqer	wqer
lname	Saeedah	Jodi	Jodeh	Boutros	Al masri	Al shanti	qwer	qwer	qwer
sex	M	M	M	F	M	M	M	M	M
blood_type	A+	B+	A+	O+	B-	B-	A+	A+	A+
bdate	2009-11-11	2011-11-11	2010-11-11	1968-04-01	1999-11-11	1990-11-11	2222-11-22	2222-11-22	2222-11-22
phone_number	12345689	12345689	12345689	23457887	12345689	12345689			
dep_name	Pediatrics	Pediatrics	Pediatrics	Surgical	Surgical	Surgical	Surgical	Surgical	Surgical
emp_id	130	132	131	133	133	133	134	134	134
address	Tulkarm	Tulkarm	Tulkarm	Jenin	Tulkarm	Jerusalem	Tulkarm	Tulkarm	Tulkarm
admission_date	2023-12-27	2023-12-27	2023-12-27	2023-12-27	2023-12-27	2023-12-27	2023-12-27	2023-12-27	2023-12-27
discharge_note	<null>	<null>	<null>	<null>	<null>	<null>	<null>	<null>	<null>
nurse_note	<null>	<null>	<null>	<null>	<null>	<null>	<null>	<null>	<null>

The Functional Dependencies are as follows:

Id_no -> fname

Id_no -> mname

Id_no -> lname

Id_no -> sex

Id_no -> blood

Id_no -> bdate

Id_no -> phone_number

Id_no -> dep_name

Id_no -> emp_id

Id_no -> address

Id_no -> admission_date

Id_no -> discharge_note

Id_no -> nurse_note

- Employee Table:

	emp_id	fname	mname	lname	sex	bdate	phone_number	salary	shift	address	pass_word
1	107	Mohammad	Ahmad	Abdo	M	1999-11-11	0595858048	1000.00	A	Tulkarm	123
2	108	Ahmad	Mohammad	Abdo	M	1999-11-11	0595858049	1000.00	B	Jenin	123
3	121	Ather	Ahmad	Haj Qassem	M	1999-11-11	0595858050	5000.00	B	Qaffin	123
4	122	Kutaiba	Ahmad	Haj Qassem	M	1999-11-11	0595858051	5000.00	C	Jerusalem	123
5	120	Ali	Ahmad	Haj Qassem	M	1999-11-11	0595858052	5000.00	A	Qaffin	123
6	109	Mahmood	Ahmad	Abdo	M	1999-11-11	0595858053	1000.00	C	Qaffin	123
7	123	Al zubair	Ahmad	Haj Qassem	M	1995-11-11	0595858044	5000.00	C	Nablus	123
8	124	Talha	Ahmad	Haj Qassem	M	1995-11-11	05958	5000.00	B	Qalqilya	123
9	125	Qassem	Ahmad	Haj Qassem	M	1995-11-11	23143	5000.00	A	Ramallah	123
10	126	Tahseen	Ahmad	Haj Qassem	M	1995-11-11	23143	5000.00	A	Ramallah	123
11	127	Tawfiq	Ahmad	Haj Qassem	M	1995-11-11	234234	5000.00	B	Anabta	123
12	130	Fatima	Mehdi	Al-Sagher	F	2000-11-20	234523	500.00	A	Tulkarm	123

The Functional Dependencies are as follows:

emp_id -> fname

emp_id -> mname

emp_id -> lname

emp_id -> sex

emp_id -> bdate

emp_id -> phone_number

emp_id -> salary

emp_id -> shift

emp_id -> address

emp_id -> pass_word

- Doctor Table:

	emp_id	dep_name	specialty	tech_id	result_desc	status	pass_word
1	121	Pediatrics		108	<null>	false	123
2	122	Pediatrics		109	<null>	false	123
3	123	Pediatrics		109	<null>	false	123
4	124	Surgical		108	<null>	false	123
5	125	Surgical		107	<null>	false	123
6	126	Surgical		107	<null>	false	123
7	127	Emergency		108	<null>	false	123
8	120	Pediatrics		107	blood = 100	• true	123

The Functional Dependencies are as follows:

emp_id -> dep_name

emp_id -> specialty

emp_id -> tech_id

emp_id -> result_desc

emp_id -> status

emp_id -> pass_word

- Nurse Table:

	emp_id	dep_name	pass_word
1	130	Pediatrics	123
2	131	Pediatrics	123
3	132	Pediatrics	123
4	133	Surgical	123
5	134	Surgical	123
6	135	Pediatrics	123
7	137	Emergency	123
8	11111	Emergency	11111

The Functional Dependencies are as follows:

emp_id -> dep_name

emp_id -> pass_word

- Lab technician Table:

	emp_id	dep_name	pass_word
1	107	Laboratory	123
2	108	Laboratory	123
3	109	Laboratory	123

The Functional Dependencies are as follows:

emp_id -> dep_name

emp_id -> pass_word

- Diagnose Table:

	id_no	doc_id	diagnosis_no	description	status
1	5001	120	6	2023/12/27--Initial diagnosis: High blo.	false

The Functional Dependencies are as follows:

diagnosis_no -> doc_id

diagnosis_no -> id_no

diagnosis_no -> description

diagnosis_no -> status

- Orders Table:

	nurse_id	doc_id	order_desc	status	order_no	patient_ID
1	131	120	2023/12/27--Take Blood Sample	• true	11	5001

The Functional Dependencies are as follows:

order_no -> patient_id

order_no -> doc_id

order_no -> nurses_id

order_no -> status

order_no -> order_desc

- Nurse_tech Table:

	nurse_id	tech_id	sample_no	pid	doc_id	sample_desc	sample_status
1	130	107	7	<null>	<null>	<null>	false
2	131	107	8	<null>	<null>	<null>	false
3	132	109	9	<null>	<null>	<null>	false
4	133	107	10	<null>	<null>	<null>	false
5	134	107	11	<null>	<null>	<null>	false
6	135	108	12	<null>	<null>	<null>	false
7	137	108	14	<null>	<null>	<null>	false
8	131	107	18	5001	120	blood	• true
9	131	107	19	5001	120	blood	• true
10	11111	107	20	<null>	<null>	<null>	false
11	<null>	108	13	<null>	<null>	<null>	false
12	<null>	109	15	<null>	<null>	<null>	false

The Functional Dependencies are as follows:

Sample_no -> nurse_id

Sample_no -> tech_id

Sample_no -> pid

Sample_no -> doc_id

Sample_no -> sample_desc

Sample_no -> sample_status

Project EER

The project's EER is then implemented as in figure 1.

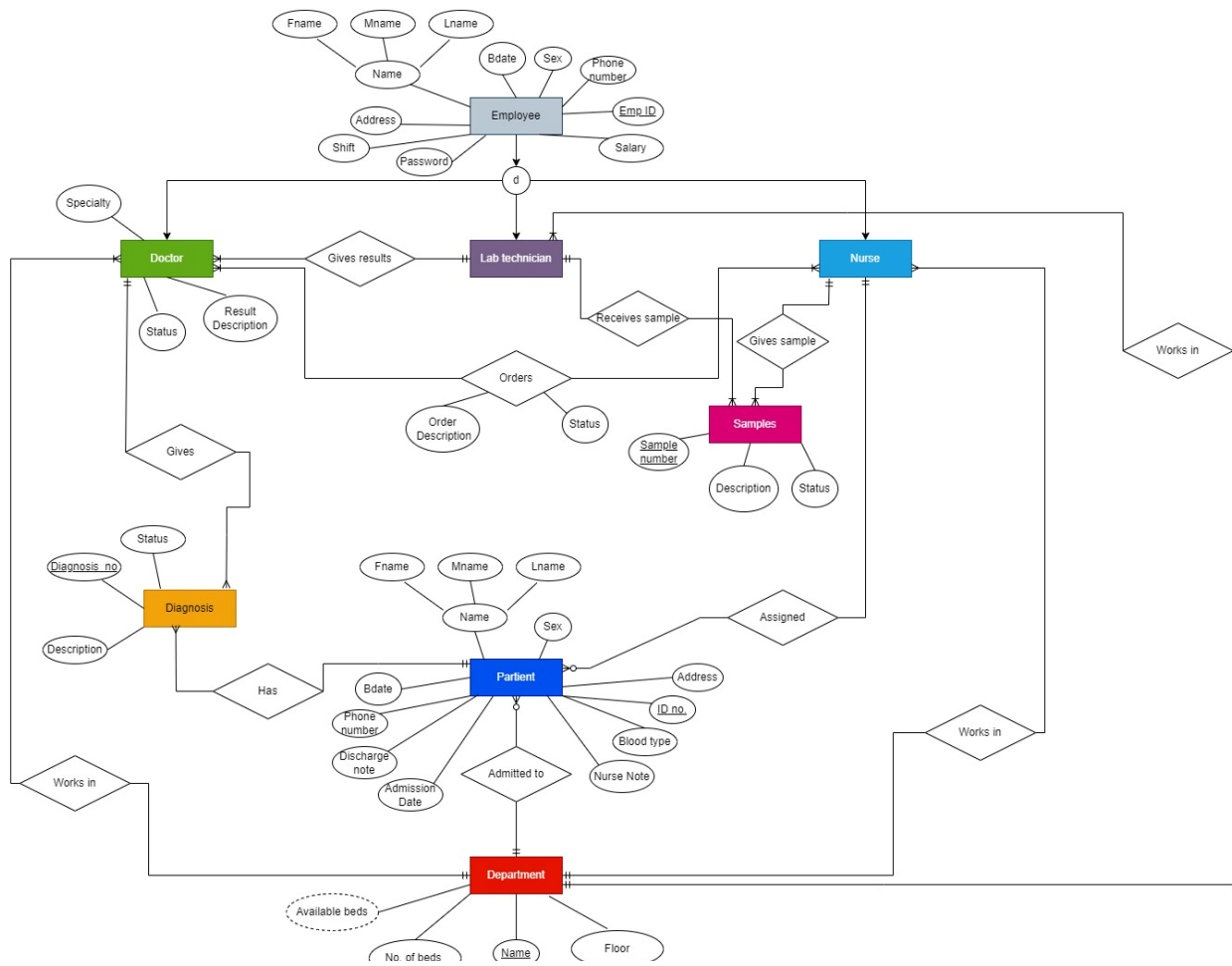


Figure 1. EER diagram for Al-Balloti Hospital

Normalization Process (BCNF)

To check if a table is in Boyce-Codd Normal Form (BCNF), we need to make sure that for every non-trivial functional dependency, the determinant is a super key. A super key is a set of one or more attributes that, taken collectively, uniquely identifies a tuple in the table. A non-trivial functional dependency is one in which the dependent attribute is not a subset of any candidate key.

- Department Table:

	dep_name	floor	no_of_beds
1	Emergency	6	6
2	Surgical	1st	6
3	Pediatrics	6	6
4	Gynecology	1st	6
5	Cardiology	2nd	6
6	ICU	1st	6
7	Laboratory	2nd	6

--Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The department table has only one candidate key (dep_name), and all non-key attributes (floor, no_of_beds) are fully functionally dependent on it. Therefore, the table is in 2NF.

--In the department table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--Both floor and no_of_beds are dependent on the whole primary key dep_name, which means this table is in BCNF because the determinant of each non-trivial functional dependency is a super key.

Since the table is in BCNF, there is no need for normalization.

- Patient Table:

[illegible]

--Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The patient table has only one candidate key (id_no), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

-- In the Patient table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--All attributes are dependent on the whole primary key id_no, which means this table is in BCNF because the determinant of each non-trivial functional dependency is a super key.

Since the table is in BCNF, there is no need for normalization.

- Employee table:

	emp_id	fname	mname	lname	sex	bdate	phone_number	salary	shift	address	pass_word
1	107	Mohammad	Ahmad	Abdo	M	1999-11-11	0595858048	1000.00	A	Tulkarm	123
2	108	Ahmad	Mohammad	Abdo	M	1999-11-11	0595858049	1000.00	B	Jenin	123
3	121	Ather	Ahmad	Haj Qassem	M	1999-11-11	0595858050	5000.00	B	Qaffin	123
4	122	Kutaiba	Ahmad	Haj Qassem	M	1999-11-11	0595858051	5000.00	C	Jerusalem	123
5	120	Ali	Ahmad	Haj Qassem	M	1999-11-11	0595858052	5000.00	A	Qaffin	123
6	109	Mahmood	Ahmad	Abdo	M	1999-11-11	0595858053	1000.00	C	Qaffin	123
7	123	Al zubair	Ahmad	Haj Qassem	M	1995-11-11	0595858044	5000.00	C	Nablus	123
8	124	Talha	Ahmad	Haj Qassem	M	1995-11-11	05958	5000.00	B	Qalqilya	123
9	125	Qassem	Ahmad	Haj Qassem	M	1995-11-11	23143	5000.00	A	Ramallah	123
10	126	Tahseen	Ahmad	Haj Qassem	M	1995-11-11	23143	5000.00	A	Ramallah	123
11	127	Tawfiq	Ahmad	Haj Qassem	M	1995-11-11	234234	5000.00	B	Anabta	123
12	130	Fatima	Mehdi	Al-Sagher	F	2000-11-20	234523	500.00	A	Tulkarm	123

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The employee table has only one candidate key (emp_id), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

--In the employee table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key

Since the table is in BCNF, there is no need for normalization.

- Doctor Table:

	emp_id	dep_name	specialty	tech_id	result_desc	status	pass_word
1	121	Pediatrics		108	<null>	false	123
2	122	Pediatrics		109	<null>	false	123
3	123	Pediatrics		109	<null>	false	123
4	124	Surgical		108	<null>	false	123
5	125	Surgical		107	<null>	false	123
6	126	Surgical		107	<null>	false	123
7	127	Emergency		108	<null>	false	123
8	120	Pediatrics		107	blood = 100	• true	123

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The Doctor table has only one candidate key (emp_id), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

--In the Doctor table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key
Since the table is in BCNF, there is no need for normalization.

- Nurse Table:

	emp_id	dep_name	pass_word
1	130	Pediatrics	123
2	131	Pediatrics	123
3	132	Pediatrics	123
4	133	Surgical	123
5	134	Surgical	123
6	135	Pediatrics	123
7	137	Emergency	123
8	11111	Emergency	11111

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The nurse table has only one candidate key (emp_id), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

-- There is no transitive dependency; each non-key attribute is directly dependent on the primary key. Therefore, the nurse table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key
Since the table is in BCNF, there is no need for normalization.

- Lab Technician Table:

	 emp_id	 dep_name	 pass_word
1	107	Laboratory	123
2	108	Laboratory	123
3	109	Laboratory	123

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

-- The lab_technician table has only one candidate key (emp_id), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

-- In the lab_technician table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key
Since the table is in BCNF, there is no need for normalization.

- Diagnose Table:

	 id_no	 doc_id	 diagnosis_no	 description	 status
1	5001	120	6	2023/12/27--Inital diagnosis: High blo.	false

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The Diagnose table has only one candidate key (diagnosis_no), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

--In the Diagnose table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key
Since the table is in BCNF, there is no need for normalization.

- Order's Table:

	nurse_id	doc_id	order_desc	status	order_no	patient_ID
1	131	120	2023/12/27--Take Blood Sample	• true	11	5001

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The orders table has only one candidate key (order_no), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

-- In the Orders table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key

Since the table is in BCNF, there is no need for normalization.

- Nurse_tech Table:

	nurse_id	tech_id	sample_no	pid	doc_id	sample_desc	sample_status
1	130	107	7	<null>	<null>	<null>	false
2	131	107	8	<null>	<null>	<null>	false
3	132	109	9	<null>	<null>	<null>	false
4	133	107	10	<null>	<null>	<null>	false
5	134	107	11	<null>	<null>	<null>	false
6	135	108	12	<null>	<null>	<null>	false
7	137	108	14	<null>	<null>	<null>	false
8	131	107	18	5001	120	blood	• true
9	131	107	19	5001	120	blood	• true
10	11111	107	20	<null>	<null>	<null>	false
11	<null>	108	13	<null>	<null>	<null>	false
12	<null>	109	15	<null>	<null>	<null>	false

-- Each column contains atomic values, and there are no repeating groups, therefore it is in 1NF.

--The Nurse_tech table has only one candidate key (sample_no), and all non-key attributes are fully functionally dependent on it. Therefore, the table is in 2NF.

---- In the Nurse_tech table, there are no transitive dependencies. Each non-key attribute is directly dependent on the primary key. Therefore, the table is in 3NF.

--The table is in BCNF because the determinant of all functional dependencies is a super key

Since the table is in BCNF, there is no need for normalization.

Project Tools

The tools utilized for this project are as follows:

- JetBrains DataGrip
- pgAdmin4
- SQL shell
- NetBeans
- Jaspersoft studio

Project GUI Implementation

the GUI opens with the login interface as in figure 2.

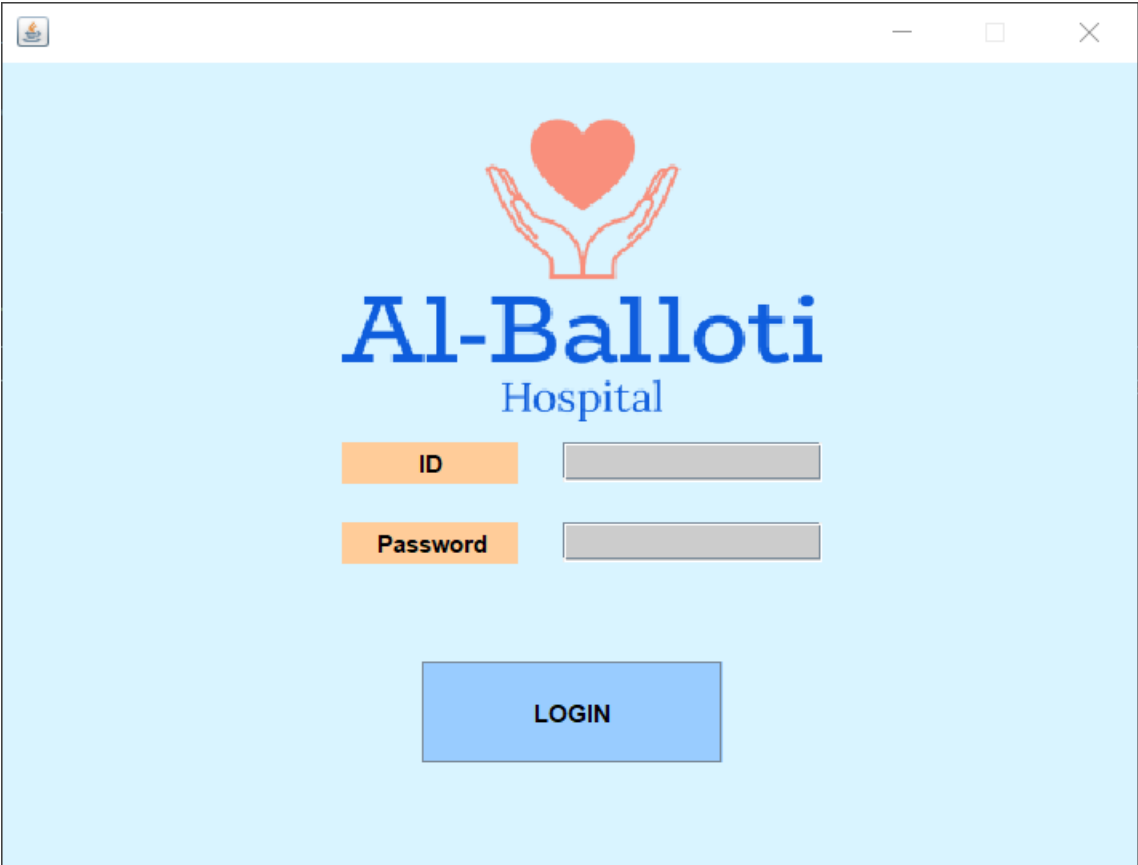
The image shows a web browser window displaying the login interface for Al-Balloti Hospital. The background is a solid light blue. At the top center, there is a logo consisting of a red heart held by two white hands. Below the logo, the text "Al-Balloti" is written in a large, bold, blue serif font, and "Hospital" is written below it in a smaller, blue serif font. Underneath the text, there are two input fields. The first field is preceded by an orange label "ID" and the second by an orange label "Password". Both labels are in a bold, black, sans-serif font. The input fields themselves are light gray with a thin black border. Below these fields is a large, light blue rectangular button with the word "LOGIN" in a bold, black, sans-serif font. The browser window has a standard title bar with a small icon on the left and minimize, maximize, and close buttons on the right.

Figure 2. Login Interface

The login interface gives access to the employees and the administrator.

Each one has separate interfaces with their different usages and permissions, which will be discussed.

1.For the Admin:

after logging in as an admin, the user faces two choices, either to Edit some data in the schema, or Add to the database as in figure 3.

As the admin presses the Add button, they will be faced with three different tabs, the Employee, Department, and Patient as seen in the figure 4, 5 and 6 respectively.

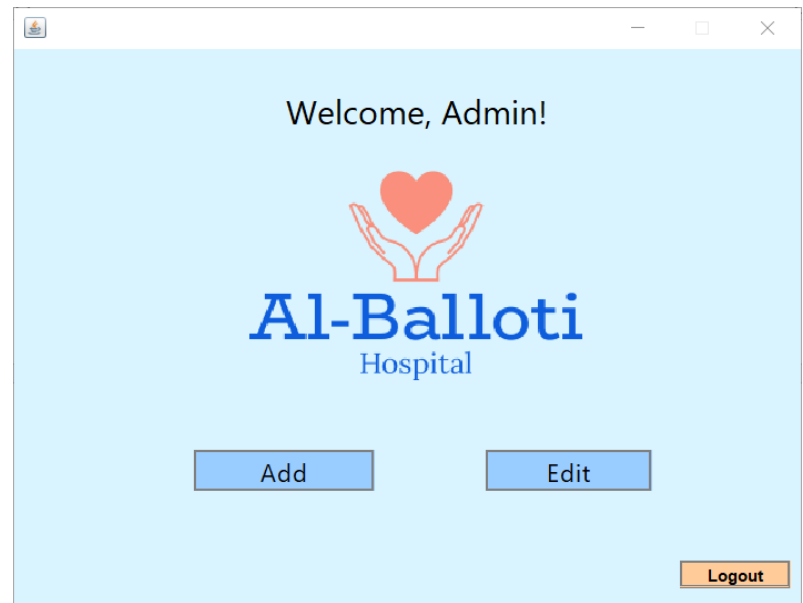


Figure 3. Admin Welcome screen

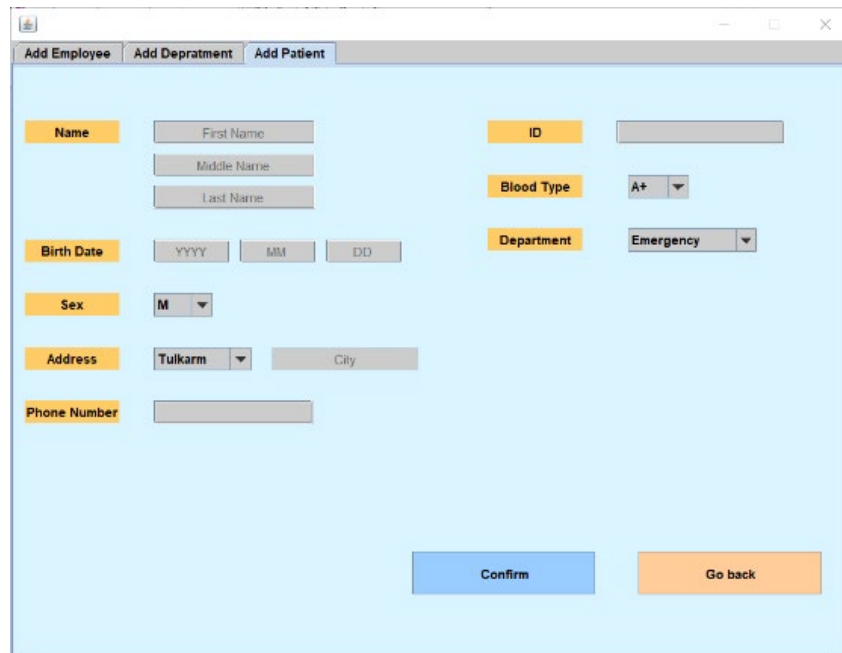
The addition interfaces for the Employee and Patient are quite similar, each filling the attributes of their corresponding entity resulting in a new row in the employee table and one of the subclasses, for the Department addition interface, it is simply adding a department name and what floor is that department at, note that the number of beds can't be edited.

The image shows a web application window titled "Employee addition interface". It has three tabs: "Add Employee", "Add Depratment", and "Add Patient". The "Add Employee" tab is active. The form contains several fields: "Employee Type" (a dropdown menu with "Doctor" selected), "Name" (three input fields for "First Name", "Middle Name", and "Last Name"), "Birth date" (three input fields for "YYYY", "MM", and "DD"), "Sex" (a dropdown menu with "M" selected), "Address" (two input fields for "Tulkarm" and "City"), "Phone Number" (one input field), "Specialty" (one input field), "ID" (one input field), "Password" (one input field), "Salary" (one input field), "Shift" (a dropdown menu with "A" selected), and "Department" (a dropdown menu with "Emergency" selected). At the bottom, there are two buttons: "Confirm" (blue) and "Go Back" (orange).

Figure 4. Employee addition interface

The image shows a web application window titled "Department Addition interface". It has three tabs: "Add Employee", "Add Depratment", and "Add Patient". The "Add Depratment" tab is active. The form contains three fields: "Name" (one input field), "Floor" (a dropdown menu with "G" selected), and "Number Of Beds" (one input field with the value "6"). At the bottom, there are two buttons: "Confirm" (blue) and "Go Back" (orange).

Figure 5. Department Addition interface



The screenshot shows a web application window titled "Patient Addition interface". It features three tabs: "Add Employee", "Add Department", and "Add Patient", with the "Add Patient" tab currently selected. The form is organized into two columns. The left column contains fields for "Name" (First Name, Middle Name, Last Name), "Birth Date" (YYYY, MM, DD), "Sex" (M), "Address" (Tulkarm, City), and "Phone Number". The right column contains fields for "ID", "Blood Type" (A+), "Department" (Emergency), and "Emergency". At the bottom right, there are two buttons: "Confirm" and "Go back".

Figure 6. Patient Addition interface

When clicking the Edit button in figure 3, an interface containing another three tabs for Employee, Department and Patient as well, as seen in figures 7, 8 and 9 respectively.

For the Employee tab seen in figure 7(a), there is a table listing the ID of the employees with their corresponding Name, the table contents can be filtered via a combo box, there's also a search bar which searches for the employee Name or ID, the search bar returns instant results while writing in its text field, this search bar will be encountered frequently in the GUI with the same functionality as we will see later on in this report.

The interface in figure 7(b), is a result of selecting an employee from the table (employee with ID 122 for example) and then clicking the Edit button in figure 7(a).

Consequently, the employee then may have their attribute values altered or entirely deleted.

Employee Department Patient

Search Ta

All Employees
All Employees
Doctor
Nurse
Lab Technician

ID	Name
122	Kutaiba Ahmad Haj Qassem
124	Talha Ahmad Haj Qassem
126	Tahseen Ahmad Haj Qassem
127	Tawfiq Ahmad Haj Qassem

Edit Go Back

Figure 7(a). Employee Edit tab

Name Kutaiba Ahmad Haj Qassem

ID 122

Password 123

Birth Date 1999 11 11

Employee Type Doctor

Sex M

Salary 5000.00

Address Jerusalem city

Shift C

Phone Number 0595858051

Department Pediatrics

Specialty

Update Delete Go Back

Figure 7(b). interface to edit a certain employee selected from the table.

for the **Department** tab seen in **figure 8(a)**, it is straight forward, there's a combo box to select the desired department, the **Update** button shows a simple interface that the user can update the name of the department or the floor which the department resides in. The button **Department Info Overview**, prints a report as shown in **figure 8(b)** containing various information relating to all departments such as the number of beds, number of patients, etc.

Employee Department Patient

Emergency

6 Beds Available

Update Department Info Overview

Go Back

Figure 8(a). Department Edit tab

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emp_count	total_salary	floor	total_beds	available_beds
0	null	2nd	6	6
3	1021110.65	G	6	6
0	null	1st	6	6
0	null	1st	6	6
3	3000.00	2nd	6	6
8	12257995.92	G	6	3

Figure 8(b). report printed after clicking Department Info Overview Button

Lastly, for the Patient tab shown in figure 9(a), the interface has the same components as in the employee tab mentioned previously, with equivalent functionality (a combo box for filtering, a search bar, and a table for data), when selecting a patient from the table and clicking Edit a similar interface to the one in figure 7(b) appears (*refer to figure 9(b)*) which lets the user edit a field or delete completely.

ID	Name	Department
5000	Hashem Muthaffar Saeed	Pediatrics
5002	Shehdeh Jodi Jodi	Pediatrics
5001	Naeem Samir Jodeh	Pediatrics
5005	Jolia abo Boutros	Surgical
5003	Vector Jaden Al masri	Surgical
5004	Hothaifah abo Al shanti	Surgical
5010	qwer wqer qwer	Surgical
5011	qwer wqer qwer	Surgical
5111	qwer wqer qwer	Surgical

Figure 9(a). Patient Edit tab.

Figure 9(b). interface to edit a certain patient selected from the table.

2.For the Employee:

Logging in as a doctor or nurse shows basically the same set of interfaces but with different permissions, like for example the diagnose button will work for the doctor but will display a warning message for the nurse, all of the permissions are based on the [project requirements](#) mentioned earlier in this report.

Now that a doctor or a nurse has logged in, the interface shows two tabs **Patients** and **Orders**, the first tab is mainly for the doctor in terms of manipulation, but the nurse can only spectate, and vice versa for the latter tab.

The Patient tab in figure 10 shows a table, search bar and a combo box all of which are of the same functionality as in the aforementioned Employee Edit tab (*refer to figure 7(a)*), after selecting a patient from the table, clicking the Manage button will show the interface in figure 11.

ID	Full Name	Blood Type
5000	Hashem Saeedah	A+
5002	Shehdeh Jodi	B+
5001	Naeem Jodeh	A+

Figure 10. Patient tab in doctor/nurse welcome interface

This interface seen in figure 11 enables the doctor to insert diagnosis to a patient, or order a nurse to do the essential vital signs for a patient, for example.

The Order Nurse button results in a simple text box letting the doctor write what they want in their own language to the nurse, the Diagnose button however, results in a frame shown in figure 12 which enables the doctor to write only the diagnosis, and lets them check if it's initial diagnosis or not.

The Application then appends the date that the diagnosis was written in, and the name of the doctor, it also attaches "initial diagnosis" if the doctor checked the initial diagnosis check box.

For the Discharge button, the Application will automatically print the discharge report for the patient followed by a discharge note by the doctor as seen in figure 16.

For the **Order** tab depicted in figure 13, logging in as a nurse, will grant the permission of the buttons shown, the tab contains the essential components talked about in the aforementioned interfaces, but has two new buttons, the Confirm Order button and the Send to Lab button, the first one shows a simple text field that the nurse must fill, it is used if the order was for simple vital signs, for example if the order is simply to take blood pressure, there is no need to send the blood pressure results to the lab, instead, that button sends the results directly to the doctor and will appear for the doctor in the "Nurse Notes" area in the patient management interface as per figure 11.

The other button is used if the order was to take a blood sample for further analysis, it will show a text field to fill as well, but it will be for the sample description that will be sent to the lab technician. **Note** that if the nurse opens the Order tab, it'll display orders of possibly multiple doctors to the nurse logged in, **but** if the doctor opens the tab, it'll display the orders that the doctor who has logged in sent possibly for multiple nurses.

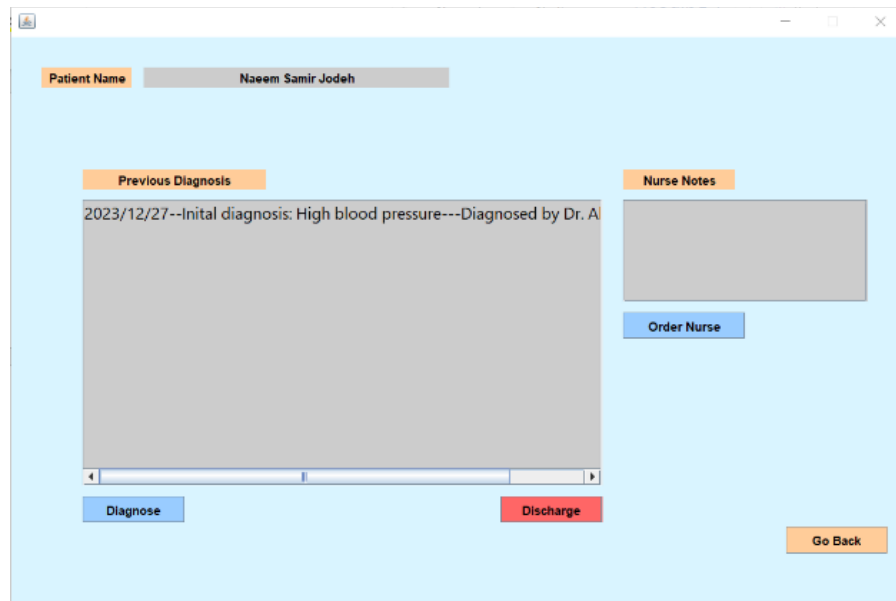


Figure 11. Manage Patient interface

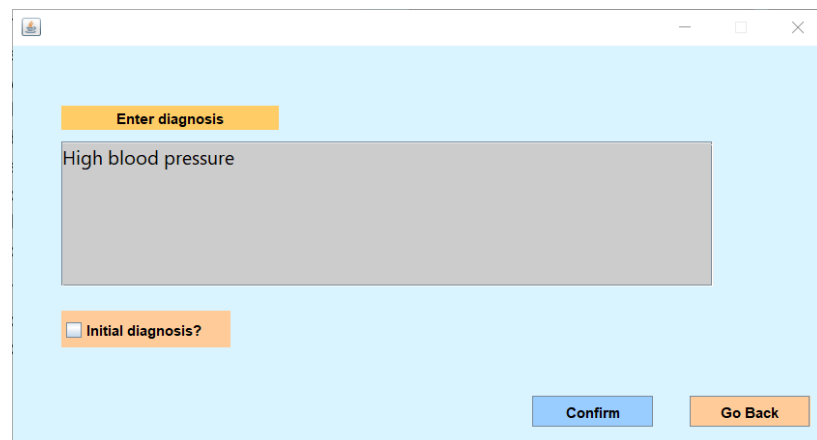


Figure 12. insert diagnosis frame

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Patient ID: **5001**

fname	mname	lname	sex	blood_type	dep_name	discharge_note	admission_date
Naeem	Samir	Jodeh	M	A+	Pediatrics	discharge note written	12/27/23, 12:00 AM

Figure 16. report of discharging a patient

Welcome, Khowailid Mehdi Al-Sagher!

Search for Order Number...

#Order	Patient Id	Diagnosis	Nurse Name	Order Discription
11	5001	Initial diagnosis: Hi...	Khowailid Al-Sagh...	2023/12/27--Take...

Confirm Order Send to Lab

Logout

Figure 13. Order tab, mainly for nurse work

Lastly, after logging in as a lab technician, an interface is shown in figure 14. The components are the same with the same functionality, there's additionally a button for submitting an order to the Doctor which prints the sample results automatically in a report depicted in figure 15.



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Date: Friday 29

Patient ID: **5001**

fname	mname	lname	sample_no	sample_desc	result_desc	emp_id
Naeem	Samir	Jodeh	19	blood	Blood test results	120

Figure 15. example report for sample results

Welcome, Mohammad Abdo!

Search Sample...

#Sample	Sample	Doctor ID	Patient ID
19	blood	120	5001

Logout Submit to Doctor

Figure 14. lab technician interface

Conclusion

In this project, it is seen that a computerized database system for a small hospital can increase productivity and shift the focus of the doctors to diagnosing patients, nurses to take care of patients, and lab technicians to not confuse orders, rather than wasting their time trying to organize pages and reports to send to the proper employee/patient.

Although this project definitely takes some load off of all of the employees, even the patients, it is surely intended for a small hospital, larger hospitals with increasing number of beds and departments and employee types may take a different way of implementing the database.

All in all, this project can be utilized in many small hospitals and preserve a lot of time for the employees which might save a lot of patients' lives along the way.

References

- Dr. Sufyan Samara slides/lectures
- [Stackoverflow.com](https://stackoverflow.com)
- [Postgresql documentation](https://www.postgresql.org/docs/)
- [DBMS lectures -Neso Academy](#)